Iris Flower Classification Using K-Nearest Neighbors (KNN)

Abstract

The objective of this project is to develop a supervised machine learning model for classifying Iris flower species. The K-Nearest Neighbors (KNN) algorithm is used to learn patterns from the dataset and predict flower categories.

Introduction

Classification is one of the most common machine learning tasks. In this project, the Iris dataset is used to train a KNN model capable of classifying flowers into three species based on their physical measurements.

Dataset Description

The Iris dataset contains 150 observations with four numerical features and three target classes: Setosa, Versicolor, and Virginica.

Methodology

1. Load Dataset
2. Split Data into Training and Testing Sets
3. Apply StandardScaler for Feature Scaling
4. Train KNN Classifier
5. Generate Predictions
6. Evaluate Model Performance

Algorithm Used

K-Nearest Neighbors (KNN) is a supervised learning algorithm that classifies data points based on the majority class of their nearest neighbors.

Evaluation Metrics

- Accuracy Score
- Confusion Matrix
- F1 Score
- Classification Report

Results

The trained KNN model successfully classified Iris flower species with high accuracy. Performance was evaluated using multiple classification metrics.

Conclusion

This project demonstrates the complete supervised machine learning workflow including data preparation, model training, prediction, and evaluation. The KNN algorithm proved effective for the Iris classification task.

Future Work

Future improvements may include comparing multiple algorithms, performing hyperparameter tuning, and applying cross-validation techniques.